SPIRE REGISTRATION INFORMATION for 2027 Spring

	PREREQUISITES	ELIGIBILITY RESTRICTIONS	CLASS NOTES/COMMENTS *INFORMATION ONLY - DOES NOT AFFECT SPIRE ENROLLMENT.*
CICS 109 INTRO TO DATA ANALYSIS IN R *Jasper McChesney*		FRESHMEN AND SOPHOMORE NON-COMPSCI MAJORS ONLY.	STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDES VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides
CICS 110 FOUNDATIONS OF PROGRAMMING *Cole Reilly, Ella Tuson*	COMPLETION OF THE R1 GEN ED (OR A SCORE OF 15 OR HIGHER ON THE MATH PLACEMENT TEST PART A), OR ONE OF THE FOLLOWING COURSES: MATH 101&102 OR MATH 104 OR MATH 127 OR MATH 128 OR MATH 131 OR MATH 132.	OPEN TO FRESHMAN, SOPHOMORE AND JUNIOR COMPSCI, INFORM, MATH, PHYSICS, COMP LINGUISTICS AND STATS & DATA SCIENCE MAJORS.	Students are not permitted to take CICS 110 after successfully completing COMPSCI 121. After eligible CS, INFORM, MATH, PHYSICS, CMPLING AND STATS & DATA SCI majors have enrolled, will open to all undergraduate freshmen, sophomore, and juniors, if available seats. Freshmen, sophomore, and juniors in other majors should wait for the eligibility restriction to be removed to self-enroll and should not request an override. Students needing special permission must request overrides via the on-line form: https://www.cics.umass.edu/academics/course-overrides.
CICS 127 INTRO TO PUBLIC INTEREST TECH *Emily Nutwell*		Open to undergraduate students.	This course counts towards the IT minor and is required for the PIT undergraduate certificate. Students needing special permission must request overrides via the on-line form: https://www.cics.umass.edu/academics/course-overrides
CICS 160 OBJECT-ORIENTED PROGRAMMING *Cole Reilly, Ella Tuson*	CICS 110 (PREVIOUSLY INFO 190S) OR COMPSCI 121 WITH A GRADE OF C OR BETTER.	UNDERGRADUATE CS, INFORM, COMP LING OR MATH MAJORS IN THE MATHCMPTNG SUBPLAN ONLY.	STUDENTS ARE NOT PERMITTED TO TAKE CICS 160 AFTER SUCCESSFULLY COMPLETING COMPSCI 186. USE OF PERSONAL LAPTOP IN CLASS IS REQUIRED. AFTER ELIGIBLE CS, INFORM, COMP LING AND MATH MAJORS IN THE MATHCMPTNG SUBPLAN HAVE ENROLLED, WILL OPEN TO OTHER MAJORS, IF AVAILABLE SEATS. ELIGIBLE NON-MAJORS SHOULD WAIT FOR THE MAJOR ELIGIBILITY RESTRICTION TO BE REMOVED TO SELF-ENROLL AND SHOULD NOT REQUEST AN OVERRIDE. STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDE VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides.
CICS 210 DATA STRUCTURES *Mordecai Golin, Marc Liberatore*	CICS 160 (PREVIOUSLY INFO 190T) WITH A GRADE OF C OR BETTER.	UNDERGRADUATE CS, INFORM, COMP LING OR MATH MAJORS IN THE MATHCMPTNG SUBPLAN ONLY.	STUDENTS ARE NOT PERMITTED TO TAKE CICS 210 AFTER SUCCESSFULLY COMPLETING COMPSCI 187. AFTER ELIGIBLE CS, INFORM COMP LING AND MATH MAJORS IN THE MATHCMPTNG SUBPLAN HAVE ENROLLED, WILL OPEN TO OTHER MAJORS, IF AVAILABLE SEATS. ELIGIBLE NON-MAJORS SHOULD WAIT FOR THE MAJOR ELIGIBILITY RESTRICTION TO BE REMOVED TO SELF-ENROLL AND SHOULD NOT REQUEST AN OVERRIDE. STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDES VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides.
CICS 256 MAKE: PHYSICAL COMPUTING *Abdul Aziz*	CICS 210 (OR COMPSCI 187) WITH A GRADE OF C OR BETTER AND BASIC MATH SKILLS (R1).	FRESHMEN AND SOPHOMORE BS-CS MAJORS ONLY.	No option for pass/fail. This course has a required lab section and counts toward the CS lab science requirement for the BS-CS. This offering is open to sophomore and freshmen BS-CS majors only. After sophomore and freshmen register, program will consider opening to junior BS-CS majors, if available seats. Junior BS-CS majors should wait for eligibility to open to self-enroll or add to the waitlist (if available) and should not request an override. Students needing special permission must request overrides via the on-line form: https://www.cics.umass.edu/academics/course-overrides.
CICS 291C S-FIND STRENGTHS/DESIGN CAREER *Casey Maloney*		CS AND INFORM SOPHOMORES AND JUNIORS ONLY.	MANDATORY PASS/FAIL. STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDES VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides
CICS 291T S-CICS TRANSFER SUCCESS *Emma Anderson*			DEPARTMENT CONSENT REQUIRED: OPEN TO INCOMING CS AND INFORM TRANSFER STUDENTS ONLY.
CICS 296P IS-PIT CLINIC *Emily Nutwell, Virginia Partridge*			INSTRUCTOR CONSENT REQUIRED. STUDENTS MUST BE ENROLLED IN THE AI4CW PROGRAM.

SPIRE REGISTRATION INFORMATION for 2027 Spring

	PREREQUISITES	ELIGIBILITY RESTRICTIONS	CLASS NOTES/COMMENTS *INFORMATION ONLY - DOES NOT AFFECT SPIRE ENROLLMENT.*
CICS 298A P-LEADRSHP:COMMUNCTNG-EXPRTISE *Emma Anderson*	CICS 160, COMPSCI 186, COMPSCI 187, OR CICS 210 WITH A GRADE OF C OR BETTER.	CS AND INFORM MAJORS ONLY.	STUDENTS IN OTHER MAJORS MAY BE CONSIDERED IF AVAILABLE SEATS. STUDENTS FROM OTHER MAJORS MAY BENEFIT FROM THIS COURSE. STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDES VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides.
CICS 305 SOC ISS IN CMPTNG *Erin Butler, Siobhan Meï, Justin Obara, Christina Sutcliffe*	CS MAJORS: ENGLWRIT 112 WITH A GRADE OF C OR BETTER (OR ENGLISH WRITING WAIVER) AND COMPSCI 220, COMPSCI 230 AND EITHER COMPSCI 240 OR COMPSCI 250. INFORM MAJORS: ENGLWRIT 112 WITH A GRADE OF C OR BETTER (OR ENGLISH WRITING WAIVER) AND INFO 248.	ODD LECS: SENIOR AND JUNIOR CS MAJORS ONLY. RESTRICTED TO CS PRIMARY MAJORS. EVEN LECS: SENIOR AND JUNIOR INFORMATICS MAJORS ONLY. RESTRICTED TO INFORMATICS PRIMARY MAJORS.	Satisfies junior year writing requirement. Odd lectures for CS majors; even lectures for INFORM majors. Eligibility limited to CS and INFORM PRIMARY majors only. CS and INFORM secondary majors must take junior year writing course within primary major department. BA-CS students are only required to complete 3 of the 4 core courses with a grade of D or better to meet eligibility. LEC 01-02 meet together; LEC 03-04..., etc. SPIRE will only recognize the ENGLWRIT112 prerequisite when taken at UMASS Amherst. Eligible majors with AP credit, transfer credit, or a waiver for ENGLWRIT112 who meet the other stated prerequisites will need assistance to enroll via the override form: https://www.cics.umass.edu/academics/course-overrides.
CICS 396A IS-DIRECTED RESEARCH GROUP *Eniola Yetunde Adegbegha*			Students must be enrolled in the ERSP program and successfully completed CICS 237 in the fall. They are then enrolled in this course in the spring of the second year.
CICS 398T P-CPT *, Jack Wileden*		UNDERGRADUATE CS AND INFORM MAJORS ONLY.	INTERNATIONAL UNDERGRADUATE STUDENTS SHOULD REFER TO THE IPO WEBSITE FOR DETAILS ABOUT CURRICULAR PRACTICAL TRAINING (CPT). SEE DETAILS ABOUT REGISTRATION AND THE CPT APPLICATION PROCESS FOR CS MAJORS: https://cicscareers.notion.site/CPT-Process-for-Internships-c0f1d14ccfe5448ba786583a49993209
CICS 598T P-CPT *Jack Wileden, Hao Zhang*		CMPSCI GRADUATE STUDENTS ONLY.	COMPUTER SCIENCE GRADUATE STUDENTS INTERESTED IN PURSUING CPT MUST SUBMIT AN EXPERIENTIAL LEARNING REQUEST IN SPIRE. STAFF WILL ENROLL GRADUATE STUDENTS FOR FALL AND SPRING IN THE UNIV SECTION AND FOR WINTER AND SUMMER IN THE UWW SECTION. DETAILS ABOUT REGISTRATION AND THE CPT APPLICATION PROCESS FOR CMPSCI GRADUATE STUDENTS: https://cicscareers.notion.site/CPT-Process-for-Internships-c0f1d14ccfe5448ba786583a49993209
COMPSCI 119 INTRO PROGRAMMING *STAFF*		FRESHMEN AND SOPHOMORE NON-CS MAJORS ONLY.	CS MAJORS ARE NOT ELIGIBLE FOR THIS COURSE. AFTER SOPHOMORE AND FRESHMEN REGISTER, WILL OPEN TO JUNIORS. IF AVAILABLE SEATS, WILL OPEN TO ALL UNDERGRADUATE NON-CS MAJORS. NON-CS JUNIORS SHOULD WAIT FOR THE ELIGIBILITY RESTRICTION TO BE REMOVED TO SELF-ENROLL AND SHOULD NOT REQUEST AN OVERRIDE. STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDES VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides.
COMPSCI 198C P-INTRO-C PROGRAMMING LANGUAGE *Meng-Chieh Chiu*	COMPSCI 121 WITH A GRADE OF B OR BETTER, OR CICS 160 (PREVIOUSLY INFO 190T, OR COMPSCI 186 OR CICS 210) WITH A GRADE OF C OR BETTER.	*FRESHMEN, SOPHOMORE, AND JUNIORS ONLY.*	STARTING FALL 2023, PASSING COMPSCI 198C IS A REQUIRED PREREQUISITE FOR COMPSCI 230. STUDENTS WITH AP OR TRANSFER CREDIT FOR COMPSCI 121 WILL NEED AN OVERRIDE FOR REGISTRATION. STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDE VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides
COMPSCI 220 PGMG METHODOLOGY *Marius Minea, James Perretta*	CICS 210 (OR COMPSCI 187) WITH A GRADE OF C OR BETTER.	CS MAJORS ONLY.	SEATS ARE BEING SAVED FOR INCOMING TRANSFER STUDENTS UNTIL THE END OF NEW STUDENT ORIENTATION AND TRANSFER (NSOT) ENROLLMENT PERIOD. REMAINING SEATS WILL BE RELEASED PRIOR TO CLASS START. REMAINING SEATS WILL BE RELEASED AT THAT TIME. STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDE VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides
COMPSCI 230 CMPTR SYS PRINC *Phuthipong Bovornkeeratiroj, Meng-Chieh Chiu*	CICS 210 (OR COMPSCI 187) WITH A GRADE OF C OR BETTER AND COMPSCI 198C.	CS MAJORS ONLY.	STARTING FALL 2023, PASSING COMPSCI 198C IS A REQUIRED PREREQUISITE. SEATS ARE BEING SAVED FOR INCOMING TRANSFER STUDENTS UNTIL THE END OF NEW STUDENT ORIENTATION AND TRANSFER (NSOT) ENROLLMENT PERIOD. REMAINING SEATS WILL BE RELEASED PRIOR TO CLASS START. STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDE VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides

SPIRE REGISTRATION INFORMATION for 2027 Spring

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COMPSCI 240 RSNG UNDER UNCERTAINTY *Chloe Becquey, Ghazaleh Parvini*	CICS 160 (PREVIOUSLY INFO 190T OR COMPSCI 187 OR CICS 210) AND MATH 132, BOTH WITH A GRADE OF C OR BETTER.	CS AND COMP LING MAJORS ONLY.	SEATS ARE BEING SAVED FOR INCOMING TRANSFER STUDENTS UNTIL THE END OF NEW STUDENT ORIENTATION AND TRANSFER (NSOT) ENROLLMENT PERIOD. REMAINING SEATS WILL BE RELEASED PRIOR TO CLASS START. STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDE VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides
COMPSCI 250 INTRO TO COMPUTATION *Mordecai Golin, Ghazaleh Parvini*	CICS 160 (PREVIOUSLY INFO 190T OR COMPSCI 187 OR ECE 241 OR CICS 210) AND MATH 132, BOTH WITH A GRADE OF C OR BETTER.	CS, CSENG, AND MATH MAJORS ONLY.	SEATS ARE BEING SAVED FOR INCOMING TRANSFER STUDENTS UNTIL THE END OF NEW STUDENT ORIENTATION AND TRANSFER (NSOT) ENROLLMENT PERIOD. REMAINING SEATS WILL BE RELEASED PRIOR TO CLASS START. STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDE VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides
COMPSCI 311 INTRO TO ALGORITHMS *Marius Minea, Ghazaleh Parvini*	CICS 210 (OR COMPSCI 187) AND COMPSCI 250 (OR MATH 455), BOTH WITH A GRADE OF C OR BETTER.	JUNIOR AND SENIOR CS MAJORS OR MATH MAJORS IN THE MATHCMPTNG SUBPLAN ONLY.	SEATS ARE BEING SAVED FOR INCOMING TRANSFER STUDENTS UNTIL THE END OF NEW STUDENT ORIENTATION AND TRANSFER (NSOT) ENROLLMENT PERIOD. AFTER JUNIOR AND SENIOR CS MAJORS OR MATH MAJORS IN THE MATHCMPTNG SUBPLAN REGISTER, WILL OPEN TO SOPHOMORES. SOPHOMORE, JUNIOR AND SENIOR CS, INFORM, COMP LING OR MATH MAJORS IN THE MATHCMPTNG SUBPLAN SHOULD WAIT FOR THE ELIGIBILITY RESTRICTION TO BE REMOVED TO SELF-ENROLL AND SHOULD NOT REQUEST AN OVERRIDE. REMAINING SEATS WILL BE RELEASED PRIOR TO CLASS START. STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDE VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides.
COMPSCI 320 INTRO TO SOFTWARE ENGINEERING *Yuriy Brun, Jaime Davila, James Perretta*	COMPSCI 220 WITH GRADE OF C OR BETTER.	JUNIOR AND SENIOR CS AND INFORM MAJORS ONLY.	No option for pass/fail. Satisfies integrative experience requirement for CS majors. Meets with COMPSCI 429. Seats are being saved for incoming transfer students until the end of new student orientation and transfer (NSOT) enrollment period. After junior and seniors register, will open to sophomores. Sophomores should wait for the eligibility restriction to be removed to self-enroll and should not request an override. Remaining seats will be released prior to class start. Students needing special permission must request override via the on-line form: https://www.cics.umass.edu/academics/course-overrides.
COMPSCI 325 INTRO TO HUMAN CMPTR INTERACTN *Ella Tuson*	CS MAJORS: CICS 210 (OR COMPSCI 187) WITH A GRADE OF C OR BETTER. INFORM MAJORS: INFO 248 AND CICS 160 (PREVIOUSLY INFO 190T OR COMPSCI 186 OR COMPSCI 187), BOTH WITH A GRADE OF C OR BETTER.	JUNIOR AND SENIOR INFORM MAJORS ONLY.	Required for INFORM majors. Will open to JR/SR CS majors after JR ramp on. Seats saved for incoming transfer students until the end of new student orientation and transfer (NSOT). Remaining seats will be released at that time. Starting S27 counts towards the Integrative Experience requirement for CS and INFORM Majors. Students needing special permission must request override via the on-line form: https://www.cics.umass.edu/academics/course-overrides.
COMPSCI 326 WEB PROGRAMMING *Timothy Richards*	COMPSCI 220 WITH A GRADE OF C OR BETTER [OR INFO 248 AND CICS 210 (OR COMPSCI 187), BOTH WITH A GRADE OF C OR BETTER].	JUNIOR AND SENIOR CS AND INFORM MAJORS ONLY.	No option for pass/fail. Satisfies integrative experience requirement for CS and Inform majors. Students needing special permission must request overrides via the on-line form: https://www.cics.umass.edu/academics/course-overrides.
COMPSCI 345 PRACT & APPLICS OF DATA MGMT *Gordon Anderson, Marco Serafini, STAFF*	CS MAJORS: CICS 210 (OR COMPSCI 187) WITH A GRADE OF C OR BETTER. INFORM MAJORS: INFO 248 AND CICS 160 (PREVIOUSLY INFO 190T OR COMPSCI 186 OR COMPSCI 187), BOTH WITH A GRADE OF C OR BETTER.	JUNIOR AND SENIOR CS AND INFORM MAJORS ONLY.	*Seats are being saved for incoming transfer students until the end of new student orientation and transfer (NSOT) enrollment period. Will open to sophomore CS and INFORM MAJORS after JR ramp on. Remaining seats will be released prior to class start. Students are expected to have basic programming skills, and be able to use laptops in class. Students who have taken COMPSCI 445 are not eligible and should not enroll. Students needing special permission must request override via the on-line form: https://www.cics.umass.edu/academics/course-overrides.*
COMPSCI 348 PRINCIPLES OF DATA SCIENCE *Gordon Anderson*	CICS 210 (OR COMPSCI 187), COMPSCI 240, AND COMPSCI 250 (OR MATH 455), ALL WITH A GRADE OF C OR BETTER.	JUNIOR AND SENIOR CS MAJORS ONLY.	STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDES VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides.
COMPSCI 349 HARNESS DATA SCI-SOCIETAL GOOD *Abhidip Bhattacharyya*	CICS 210 (OR COMPSCI 187) AND (COMPSCI 220 OR COMPSCI 230 OR INFO 248) AND (COMPSCI 240 OR STATISTC 240 OR PSYCH 240 OR OIM 240 OR RES-ECON 212 OR SOCIOL 212 OR STATISTC 315), ALL WITH A GRADE OF C OR BETTER.	JUNIOR AND SENIOR CS AND INFORM MAJORS ONLY.	NO PASS/FAIL OPTION. STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDES VIA ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides

SPIRE REGISTRATION INFORMATION for 2027 Spring

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COMPSCI 370 INTRO TO COMPUTER VISION *Mustafa Chasmai, Logan Lawrence*	COMPSCI 240 OR COMPSCI 383 WITH A GRADE OF C OR BETTER.	JUNIOR AND SENIOR CS MAJORS ONLY.	A BACKGROUND IN LINEAR ALGEBRA (MATH 235) IS RECOMMENDED. STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDES VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides
COMPSCI 377 OPERATING SYSTEMS *Meng-Chieh Chiu*	COMPSCI 230 WITH A GRADE OF C OR BETTER.	JUNIOR AND SENIOR CS MAJORS ONLY.	STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDES VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides
COMPSCI 383 ARTIFICIAL INTELLIGENCE *Abhidip Bhattacharyya*	CS MAJORS: (CICS 210 OR COMPSCI 187) AND COMPSCI 240 (OR STATISTC 315; PREVIOUSLY STATISTC 515), BOTH WITH A GRADE OF C OR BETTER. INFORM MAJORS: INFO 348 AND STATISTC 315, BOTH WITH A GRADE OF C OR BETTER.	JUNIOR AND SENIOR CS AND INFORM MAJORS ONLY.	SOME KNOWLEDGE OF PYTHON IS ASSUMED FOR ELIGIBLE STUDENTS TO USE COMPSCI 187 IN PLACE OF THE CICS 210 PREREQUISITE. STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDES VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides
COMPSCI 389 INTRO TO MACHINE LEARNING *Philip Thomas*	CICS 210 (OR COMPSCI 187) AND (COMPSCI 240 OR STATISTC 315 OR STATISTC 240 OR PSYCH 240 OR OIM 240 OR RES-ECON 212 OR SOCIOL 212) AND MATH 132, ALL WITH A GRADE OF C OR BETTER.	JUNIOR AND SENIOR CS AND INFORM MAJORS ONLY.	STUDENTS WHO HAVE COMPLETED COMPSCI 589 ARE NOT ELIGIBLE FOR THIS COURSE AND SHOULD NOT ENROLL. STUDENTS WITH PRIOR EXPERIENCE IN MACHINE LEARNING OR WHO ARE ALREADY PASSIONATE ABOUT THE SUBJECT ARE ENCOURAGED TO TAKE COMPSCI 589 INSTEAD. ELIGIBLE STUDENTS WHO HAVE COMPLETED COMPSCI 187 WITH A C OR BETTER MAY USE IT IN PLACE OF CICS 210; DO NOT REQUEST AN OVERRIDE, BUT WAIT FOR THE CHANGE TO SHOW ON SPIRE TO SELF-ENROLL. STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDES VIA ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides
COMPSCI 398A P-CREDITED INTERNSHIP			CICS CAREERS AND INSTRUCTOR PERMISSION REQUIRED.
COMPSCI 412 QUANTUM INFORMATION SCIENCE *Filip Rozpedek*	MATH 132 AND MATH 235 AND COMPSCI 240 (OR STATISTC 315 OR PHYSICS 281 OR PHYSICS 287), ALL WITH A GRADE OF C OR BETTER.	JUNIOR AND SENIOR CS AND INFORM MAJORS ONLY.	NOT ON THE LIST OF PRE-APPROVED INFORM ELECTIVES. AFTER JUNIOR AND SENIOR CS AND INFORM MAJORS HAVE ENROLLED, THIS COURSE WILL OPEN TO JUNIOR AND SENIOR PHYSICS, MATH, AND ENGINEERING MAJORS, WHO SHOULD WAIT FOR THE COURSE TO OPEN TO SELF-ENROLL. MATH 233, COMPSCI 250 AND 311 ARE HELPFUL BUT NOT NECESSARY. STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDES VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides
COMPSCI 420 SOFTWARE ENTREPRENEURSHIP *Bogdan Prokopovych*	COMPSCI 320 (OR COMPSCI 325 OR COMPSCI 326) WITH A GRADE OF C OR BETTER.	JUNIOR AND SENIOR CS AND INFORM MAJORS ONLY.	THIS COURSE QUALIFIES AS A 'CAPSTONE COURSE' FOR THE UNDERGRADUATE CERTIFICATE IN PUBLIC INTEREST TECHNOLOGY, ASSUMING THE STUDENT WORK PRODUCED IS DEEMED SUITABLE BY THE INSTRUCTOR. MAKE SURE TO DISCUSS USING THIS COURSE FOR THIS PURPOSE WITH THE INSTRUCTOR AT THE BEGINNING OF THE COURSE. FOR MORE INFORMATION ON THE CERTIFICATE: https://groups.cs.umass.edu/pit/undergraduate-pit-certificate/. STUDENTS ENROLLED IN THE UNDERGRADUATE CERTIFICATE IN PUBLIC INTEREST TECHNOLOGY ARE ELIGIBLE VIA OVERRIDE. IF YOU HAVE QUESTIONS ABOUT THE PIT CERTIFICATE, EMAIL THE PIT CERTIFICATE PROGRAM DIRECTOR, PROFESSOR CHARLIE SCHWEIK AT cschweik@umass.edu. STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDES VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides
COMPSCI 426 SCALABLE WEB SYSTEMS *Timothy Richards*	COMPSCI 320 OR COMPSCI 326 WITH A GRADE OF C OR BETTER.	JUNIOR AND SENIOR CS AND INFORM MAJORS ONLY.	It is recommended to have completed Compsci 220 (programming methodology) and Compsci 230 (computer systems principles), or Compsci 377 (operating systems), however, this is not required but will make your experience more rich (e.g., background in web app deployment, basic security, network protocols, processes, client server architecture, and general operating systems concepts). Students needing special permission must request overrides via the on-line form: https://www.cics.umass.edu/academics/course-overrides
COMPSCI 429 SOFTWARE ENGINEERING PROJ MGMT *Yuriy Brun, Jaime Davila, James Perretta*	COMPSCI 320 WITH A GRADE OF B OR BETTER.	UNDERGRADUATE CS MAJORS ONLY.	Instructor consent required. Meets with COMPSCI 320. Eligible students should complete the instructor's review form: https://forms.gle/yawQeTvjvqTtAxNv6. Once approved to enroll, refer to instructor permission on the override request and submit via the on-line form: https://www.cics.umass.edu/academics/course-overrides.

SPIRE REGISTRATION INFORMATION for 2027 Spring

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COMPSCI 445 INFORMATION SYSTEMS *Marco Serafini*	COMPSCI 220 (OR COMPSCI 230) AND COMPSCI 311 AND COMPSCI 345, ALL WITH GRADE OF C OR BETTER.	JUNIOR AND SENIOR CS MAJORS ONLY.	WITH INSTRUCTOR PERMISSION, STUDENTS WITH PRIOR EXPERIENCE IN DATABASES MAY SUBSTITUTE ONE OF COMPSCI 320, 326, OR 377 WITH A GRADE OF C OR BETTER FOR THE COMPSCI 345 PREREQUISITE. IF AVAILABLE SEATS, STUDENTS WHO ARE NOT CS MAJORS BUT HAVE SUFFICIENT PROGRAMMING EXPERIENCE OR CS GRADUATE STUDENTS MAY SUBMIT AN OVERRIDE FOR INSTRUCTOR CONSIDERATION. STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDE VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides
COMPSCI 453 COMPUTER NETWORKS *Arun Venkataramani*	COMPSCI 230 (OR COMPSCI 377) WITH A GRADE OF C OR BETTER.	JUNIOR AND SENIOR CS MAJORS ONLY.	STUDENTS WILL BENEFIT SIGNIFICANTLY FROM A PRIOR OR CONCURRENT EXPOSURE TO PROBABILITY (COMPSCI 240), ALGORITHMS (COMPSCI 311), AND MULTI-THREADED PROGRAMMING (COMPSCI 377 OPERATING SYSTEMS) ALTHOUGH THESE ARE NOT STRICT PREREQUISITES, STUDENTS MUST BE ABLE TO PROGRAM IN A STRUCTURED HIGH-LEVEL PROGRAMMING LANGUAGE, SUCH AS JAVA, C OR C++, PYTHON, ETC. PROGRAMMING PROJECTS IN A LANGUAGE OF STUDENT CHOICE. STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDES VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides
COMPSCI 466 APPLIED CRYPTOGRAPHY *Adam O'Neill*	COMPSCI 311 WITH A GRADE OF C OR BETTER.	JUNIOR AND SENIOR CS MAJORS ONLY.	STUDENTS WHO HAVE TAKEN HIGHER-LEVEL MATHEMATICS (IN PARTICULAR MATH 411, MATH 455, MATH 471, OR MATH 513/COMPSCI 575) ARE PARTICULARLY INVITED TO ENROLL. STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDES VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides
COMPSCI 485 APPLICATIONS OF NLP *Katrin Erk*	COMPSCI 220 AND COMPSCI 240, BOTH WITH A GRADE OF C OR BETTER (OR LINGUIST 429H/492B WITH A GRADE OF C OR BETTER).	JUNIOR AND SENIOR CS AND COMP LING MAJORS ONLY.	PREVIOUSLY COMPSCI 490A. STUDENTS FROM OTHER DISCIPLINES WILL BE CONSIDERED AS WELL IF AVAILABLE SEATS. STUDENTS NEEDING INSTRUCTOR PERMISSION MUST REQUEST OVERRIDES VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides
COMPSCI 490M MODERN ALGORITHM ANALYSIS *Andrew McGregor*	COMPSCI 240 (OR STATISTCS 315) AND COMPSCI 311, BOTH WITH A GRADE OF B+ OR BETTER (OR COMPSCI 240, STATISTCS 315/515, COMPSCI 311, MATH 233, AND MATH 235, ALL WITH A GRADE OF C OR BETTER).	JUNIOR AND SENIOR CS AND INFORM MAJORS ONLY.	Recommended for students who enjoyed COMPSCI 311 and the related mathematics who want to learn more advanced algorithmic topics. Students needing special permission must request overrides via the on-line form: https://www.cics.umass.edu/academics/course-overrides
COMPSCI 491G S-COMPUTER NETWORKING LAB *Phuthipong Bovornkeeratiroj*	COMPSCI 453 WITH A GRADE OF C OR BETTER.	JUNIOR AND SENIOR CS MAJORS ONLY.	STUDENTS MAY WORK INDEPENDENTLY IN LGRT 224 TO COMPLETE LAB ASSIGNMENTS. STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDES VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides
COMPSCI 496C IS-SOC ENTRPRNRSHP LAUNCHPAD *Neena Thota*	COMPSCI 420 WITH A GRADE OF C OR BETTER.	JUNIOR AND SENIOR CS AND INFORM MAJORS ONLY.	INSTRUCTOR CONSENT REQUIRED. OPEN TO CS AND INFORM MAJORS WHO HAVE SUCCESSFULLY COMPLETED EITHER COMPSCI 420 OR HAVE EQUIVALENT ACADEMIC OR INDUSTRY EXPERIENCE.
COMPSCI 501 FORMAL LANGUAGE THEORY *David Barrington*	UNDERGRADUATES: COMPSCI 311 WITH A GRADE OF C OR BETTER.	LEC 01: JUNIOR AND SENIORS ONLY. LEC 02: MS-CMPSCI GRADUATE STUDENTS ONLY.	LEC 01 FOR UNDERGRADS; LEC 02 FOR GRADS. IT IS RECOMMENDED THAT UNDERGRADUATE STUDENTS HAVE A GRADE OF B- OR BETTER IN COMPSCI 311 IN ORDER TO ATTEMPT 501. INSTRUCTOR WILL CONSIDER CONCURRENT ENROLLMENT WITH COMPSCI 311 FOR STUDENTS WHO HAVE COMPLETED COMPSCI 250 WITH A GRADE OF A, IF AVAILABLE SEATS. SEATS HELD IN LECT 02 FOR INCOMING GRAD STUDENT REGISTRATION. STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDES VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides
COMPSCI 508 ETHICAL CONSIDRTNS IN DS&AI *Michelle Trim*	UNDERGRADUATES: CICS 305 WITH A GRADE OF C OR BETTER.	LEC 01: JUNIOR AND SENIOR CS AND INFORM MAJORS ONLY. LEC 02: CMPSCI GRADUATE STUDENTS ONLY.	LEC 01 FOR UNDERGRADS; LEC 02 FOR GRADS. THIS COURSE FULFILLS THE ETHICS REQUIREMENT FOR HLS. STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDES VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides

SPIRE REGISTRATION INFORMATION for 2027 Spring

	PREREQUISITES	ELIGIBILITY RESTRICTIONS	CLASS NOTES/COMMENTS *INFORMATION ONLY - DOES NOT AFFECT SPIRE ENROLLMENT.*
COMPSCI 514 ALGORITHMS FOR DATA SCIENCE *Mingda Qiao*	UNDERGRADUATES: COMPSCI 240 (OR STATISTCS 315) AND COMPSCI 311, BOTH WITH A GRADE OF B+ OR BETTER (OR COMPSCI 240, STATISTCS 315, COMPSCI 311, MATH 233, AND MATH 235, ALL WITH A GRADE OF C OR BETTER).	LEC 01: JUNIOR AND SENIOR CS MAJORS ONLY. LEC 02: MS-CMPSCI GRADUATE STUDENTS ONLY.	LEC 01 FOR UNDERGRADS; LEC 02 FOR GRADS. SEATS HELD IN LEC 02 FOR INCOMING GRAD STUDENT REGISTRATION. STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDES VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides
COMPSCI 520 THEORY AND PRACTICE-SOFTENGIN *Heather Conboy*	UNDERGRADUATES: COMPSCI 320 (OR COMPSCI 220 AND COMPSCI 326) WITH A GRADE OF C OR BETTER.	LEC 01: JUNIOR AND SENIOR CS MAJORS ONLY. LEC 02: MS-CMPSCI GRADUATE STUDENTS ONLY.	LEC 01 for undergrads; LEC 02 for grads. Seats held in LEC 02 for incoming grad student registration. Students needing special permission must request overrides via the on-line form: https://www.cics.umass.edu/academics/course-overrides.
COMPSCI 532 SYSTEMS FOR DATA SCIENCE *Peter Klemperer*	UNDERGRADUATES: COMPSCI 377 AND COMPSCI 445, BOTH WITH A GRADE OF C OR BETTER.	LEC 01: JUNIOR AND SENIOR CS MAJORS ONLY. LEC 02: MS-CMPSCI GRADUATE STUDENTS ONLY. C1 LEC 01: ONLINE MS-CMPSCI PROGRAM STUDENTS ONLY.	LEC 01 for undergrads; LEC 02 for eligible MS-CMPSCI grads; seats held in LEC 02 for incoming student registration. C1 LEC 01 for online MS-CMPSCI program students only. Course is online with optional synchronous live lectures on MW 4:00-5:15 PM EDT. All exams will be in-person for all students who are not enrolled in the online MS-CMPSCI program. PhDs or any student with a tuition waiver cannot enroll in C1 LEC 01. PhD or other graduate level students who need a university course should register in LEC 02. Students needing special permission must request overrides via the on-line form: https://www.cics.umass.edu/academics/course-overrides.
COMPSCI 535 COMPUTER ARCHITECTURE *Charles Weems*	UNDERGRADUATES: COMPSCI 335 WITH A GRADE OF C OR BETTER.	LEC 01: JUNIOR AND SENIOR CS MAJORS ONLY. LEC 02: MS-CMPSCI GRADUATE STUDENTS ONLY.	LEC 01 FOR UNDERGRADS; LEC 02 FOR GRADS. SEATS HELD IN LEC 02 FOR INCOMING GRAD STUDENT REGISTRATION. STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDES VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides
COMPSCI 561 SYSTEM DEFENSE AND TEST *Lurene Grenier*	UNDERGRADUATES: COMPSCI 360 (OR COMPSCI 460 OR COMPSCI 560 OR COMPSCI 660 OR E&C-ENG 371) AND COMPSCI 453 (OR E&C-ENG 325), BOTH WITH A GRADE OF C OR BETTER.	LEC 01: JUNIOR AND SENIOR CS MAJORS ONLY. LEC 02: GRADUATE CMPSCI AND ECE STUDENTS ONLY. C1 LEC 01: GRADUATE CMPSCI STUDENTS ONLY.	LEC 01 for undergrads; LEC 02 for grads. If available seats, eligible CSENG majors with ECE 371 and ECE 325 must wait for course to open to self-enroll (do not request overrides). C1 LEC 01: online course for graduate CMPSCI students only. Graduate students from other departments with instructor permission, if available seats. Students needing special permission must request overrides via the on-line form: https://www.cics.umass.edu/academics/course-overrides.
COMPSCI 563 INTERNET LAW AND POLICY *Marvin Cable*	UNDERGRADUATES: COMPSCI 311 OR COMPSCI 383 OR COMPSCI 360 (PREVIOUSLY 460) WITH C OR BETTER.	LEC 01: JUNIOR AND SENIOR CS MAJORS ONLY. LEC 02: CMPSCI AND ECE GRADUATE STUDENTS ONLY. C1 LEC 01: MS-CMPSCI GRADUATE STUDENTS ONLY.	UNIV LEC 01 FOR UNDERGRADS; UNIV LEC 02 FOR ELIGIBLE GRADS; SEATS HELD IN LEC 02 FOR INCOMING STUDENT REGISTRATION. U+ LEC 01 SYNCHRONOUS REMOTE ATTENDANCE AT SPECIFIED MEETING TIMES IS REQUIRED; FOR MS-CMPSCI STUDENTS AND OTHER GRADUATE STUDENTS (EXCEPT MS-ECE) WITH INSTRUCTOR PERMISSION. PHDS OR ANY STUDENT WITH A TUITION WAIVER CANNOT ENROLL IN A U+ SECTION. STUDENTS NEEDING SPECIAL PERMISSION FOR EITHER UNIVERSITY SECTION MUST REQUEST OVERRIDES VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides.
COMPSCI 564 CYBER EFFECTS *Jeffrey Hamalainen, Seth Landsman, Nick Merlino, Edward Walters, Adam Woodbury*	UNDERGRADUATES: COMPSCI 230 (OR E&C-ENG 322) AND COMPSCI 360 (PREVIOUSLY COMPSCI 460 OR COMPSCI 365 OR COMPSCI 367 (PREVIOUSLY COMPSCI 390R) OR COMPSCI 466 OR E&C-ENG 371), BOTH WITH A GRADE OF C OR BETTER.	LEC 01: JUNIORS AND SENIORS ONLY. LEC 02: GRADUATE CMPSCI AND ECE STUDENTS ONLY. C1 LEC 01: GRADUATE CMPSCI STUDENTS ONLY.	*LEC 01 for undergrads; LEC 02 for grads. Undergrads may substitute any introduction to computer/network security course at the 300-level or higher with a grade of C or better for COMPSCI 360, with permission of instructor. Seats held in LEC 02 for incoming student registration. Graduate students from other departments with instructor permission, if available seats. C1 LEC 01: Graduate CMPSCI students only. Other graduate students may be considered with instructor permission. Synchronously remote attendance via zoom required. Students should attend all lecture sessions. Students needing special permission must request overrides via the on-line form: https://www.cics.umass.edu/academics/course-overrides.*
COMPSCI 589 MACHINE LEARNING *Bruno Castro da Silva*	UNDERGRADUATES: [MATH 545 AND STATISTCS 315 (PREVIOUSLY STATISTCS 515) AND COMPSCI 240, ALL WITH A GRADE OF C OR BETTER.] OR [MATH 545 WITH A GRADE OF C OR BETTER AND COMPSCI 240 WITH A GRADE OF B+ OR BETTER.] OR [MATH 235 AND MATH 233, BOTH WITH A GRADE OF B+ OR BETTER, AND STATISTCS 315 (PREVIOUSLY STATISTCS 515) AND COMPSCI 240, BOTH WITH A GRADE OF C OR BETTER.] OR [MATH 235 AND MATH 233 AND COMPSCI 240, ALL WITH A GRADE OF B+ OR BETTER.]	LEC 01: JUNIOR AND SENIOR CS MAJORS ONLY. LEC 02: MS-CMPSCI GRADUATE STUDENTS ONLY. C1 LEC 01: ONLINE MS-CMPSCI GRADUATE STUDENTS ONLY.	*LEC 01 for undergrads; LEC 02 for grads. Seats held in LEC 02 for incoming grad student registration. C1 LEC 01 online MS-CMPSCI students and MS-STATS (Newton) only. Students needing special permission must request overrides via the on-line form: https://www.cics.umass.edu/academics/course-overrides.*

SPIRE REGISTRATION INFORMATION for 2027 Spring

	PREREQUISITES	ELIGIBILITY RESTRICTIONS	CLASS NOTES/COMMENTS *INFORMATION ONLY - DOES NOT AFFECT SPIRE ENROLLMENT.*
COMPSCI 590L MAKING PREDICTIONS *Justin Domke*	*UNDERGRADUATES: COMPSCI 383 WITH A GRADE OF C OR BETTER.*	*LEC 01: JUNIOR AND SENIOR CS MAJORS ONLY.* *LEC 02: MS-CMPSCI GRAD STUDENTS ONLY.*	*No formal prerequisites. Informally, students should be familiar with calculus (E.G. MATH 131, 132) and basic probability theory. LEC 01 for undergrads; LEC 02 FOR grads. Seats held in LEC 02 for incoming grad students. Students needing instructor permission must request overrides via the on-line form: https://www.cics.umass.edu/academics/course-overrides.*
COMPSCI 590NN NEURAL NETWRKS-AI AND NEUROSCI *Hava Siegelmann*	UNDERGRADUATES: COMPSCI 311 AND COMPSCI 389 (OR COMPSCI 589), BOTH WITH A GRADE OF C OR BETTER.	LEC 01: JUNIOR AND SENIOR CS MAJORS ONLY. LEC 02: GRADUATE MS-CMPSCI STUDENTS ONLY.	Meets with COMPSCI 690NN. Lec 01 is for Undergrads; Lec 02 is for Grads; seats held for incoming graduate student registration. COMPSCI 311 is a required prerequisite for undergraduates, effective Spring 2027. Background in AI and Machine Learning as well as practical knowledge in Neural Networks. Students may only take this class once, either 590NN or 690NN, which are primarily differentiated through the quantity and difficulty of homework and projects, insofar as 690NN carries a higher level of academic rigor. Students needing special permission must request overrides via the on-line form: https://www.cics.umass.edu/academics/course-overrides.
COMPSCI 590QC QUANTUM COMMUNICATION *Gayane Vardoyan*	UNDERGRADUATES: MATH 235 AND COMPSCI 240 (OR STATISTC 315/515), BOTH WITH A GRADE OF C OR BETTER.	LEC 01: JUNIOR AND SENIOR CS MAJORS ONLY. LEC 02: MS-CMPSCI STUDENTS ONLY.	MEETS WITH COMPSCI 690QC. LEC 01 FOR UNDERGRADS; LEC 02 FOR GRADS. STUDENTS CAN ONLY RECEIVE CREDIT FOR EITHER 590QC OR 690QC, NOT BOTH, REGARDLESS OF SEMESTER. SWAPPING BETWEEN 590QC AND 690QC IS ONLY CONSIDERED BEFORE THE END OF ADD/DROP. SEATS HELD FOR INCOMING GRAD STUDENT REGISTRATION. FAMILIARITY WITH PROBABILITY THEORY, LINEAR ALGEBRA, AND COMPLEX NUMBERS. NO PRIOR FAMILIARITY WITH QUANTUM CONCEPTS NEEDED. PHYSICS STUDENTS SHOULD WAIT UNTIL OPEN TO SELF-ENROLL. ALTHOUGH NOT ON THE LIST OF PRE-APPROVED INFORM ELECTIVES, INTERESTED JUNIOR AND SENIOR INFORM MAJORS MEETING PREREQUISITE WILL BE CONSIDERED VIA OVERRIDE. STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDES VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides
COMPSCI 590RM RESEARCH METHODS-EMPIRICAL CS *David Jensen*		LEC 01: JUNIOR AND SENIOR CS AND INFORM MAJORS IN COMMONWEALTH HONORS COLLEGE. LEC 02: GRADUATE CMPSCI STUDENTS ONLY.	Meets with COMPSCI 602. Department consent required for LEC 01: Undergraduate CHC CS and INFORM majors with HPD and instructor permission. For undergraduates, this course can be used to satisfy the 499Y requirement for departmental honors (HN-DH). For general study & multidisciplinary honors (HN-GSMH), students whose theses or projects have a substantial empirical component, need to obtain CHC permission to satisfy the 499Y requirement before submitting override request. Does not count as a CS elective for the CS major (BA or BS). Students needing special permission must request overrides via the on-line form: https://www.cics.umass.edu/academics/course-overrides.
COMPSCI 590X DECARBONIZATION AND DATA SCI *Jayant Taneja*	UNDERGRADUATES: EITHER [COMPSCI 240 (OR STATISTC 515) AND CICS 210 (OR COMPSCI 187) BOTH WITH A GRADE OF C OR BETTER] OR [INFO 348 WITH A GRADE OF C OR BETTER].	LEC 01: JUNIOR AND SENIOR CS AND INFORM MAJORS ONLY. LEC 02: MS-CMPSCI STUDENTS ONLY.	LEC 01 for undergrads; LEC 02 for grads. Seats held for incoming graduate student registration. Students needing special permission must request overrides via the on-line form: https://www.cics.umass.edu/academics/course-overrides.
COMPSCI 601 COMPUTATION THEORY *Hung Le*		CMPSCI GRADUATE STUDENTS ONLY.	PREREQUISITES: AN UNDERGRADUATE COURSE IN AUTOMATA THEORY AND FORMAL LANGUAGES SUCH AS COMPSCI 501 OR PERMISSION OF INSTRUCTOR. INSTRUCTOR PERMISSION REQUIRED FOR UNDERGRADUATE CS MAJORS CLEARED BY THE UPD, IF AVAILABLE SEATS. SEATS HELD FOR INCOMING GRAD STUDENT REGISTRATION. STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDES VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides
COMPSCI 602 RESEARCH METHODS-EMPIRICAL CS *David Jensen*		CS PhD students only.	Meets with COMPSCI 590RM. Open to CS PhD students only. Seats held for incoming graduate registration. Undergraduate CHC CS majors with instructor, HPD, and UPD permission. Students needing special permission must request overrides via the on-line form: https://www.cics.umass.edu/academics/course-overrides.
COMPSCI 611 ADVANCED ALGORITHMS *Hung Le, Andrew McGregor*		CMPSCI GRADUATE STUDENTS ONLY.	MATHEMATICAL MATURITY WITH KNOWLEDGE OF ALGORITHMS AT THE LEVEL OF COMPSCI 311 EXPECTED OF INCOMING CS GRADUATE STUDENTS. INSTRUCTOR PERMISSION REQUIRED FOR UNDERGRADUATE CS MAJORS CLEARED BY THE UPD, IF AVAILABLE SEATS. SEATS HELD FOR INCOMING GRAD STUDENT REGISTRATION. STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDES VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides.

SPIRE REGISTRATION INFORMATION for 2027 Spring

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COMPSCI 614 RANDMZD ALGRTHMS W/APPLC TO DS *Cameron Musco*		CMPSCI GRADUATE STUDENTS ONLY.	STRONG BACKGROUND IN ALGORITHMS AND MATHEMATICS, PARTICULARLY IN LINEAR ALGEBRA AND PROBABILITY IS REQUIRED. NOT RECOMMENDED FOR UNDERGRADUATES AND MS STUDENTS UNLESS SUCCESSFUL COMPLETION OF EITHER COMPSCI 514 OR COMPSCI 611. IF UNSURE OF PREPARATION, CONTACT INSTRUCTOR. STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDES VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides.
COMPSCI 620 ADV S/WENG:SYNTHESIS *Madeline Endres*		CMPSCI GRADUATE STUDENTS ONLY.	Prerequisite of COMPSCI 320 with a grade of C or better required for seniors, or permission of instructor. Seats held for incoming grad student registration. Students needing special permission must request overrides via the on-line form: https://www.cics.umass.edu/academics/course-overrides.
COMPSCI 621 ADV SOFTENG:ANALYSIS & EVAL *Juan Zhai*		GRADUATE CMPSCI STUDENTS ONLY.	Seats held for incoming student registration. Students needing special permission must request overrides via the on-line form: https://www.cics.umass.edu/academics/course-overrides.
COMPSCI 627 FIXING PLATFORM POWER *Ethan Zuckerman*		CMPSCI GRADUATE STUDENTS ONLY.	CROSS-LISTED WITH COMM 627 AND SPP 627.STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDES VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides
COMPSCI 645 DB DSGN & IMPLMNTN *Alexandra Meliou*		CMPSCI GRADUATE STUDENTS ONLY.	STUDENTS SHOULD HAVE TAKEN AN UNDERGRADUATE LEVEL COURSE ON OPERATING SYSTEMS (COMPSCI 377) OR DATABASES (COMPSCI 445). SEATS HELD FOR INCOMING GRAD STUDENT REGISTRATION. STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDES VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides
COMPSCI 651 OPTIMIZATION IN CS *Madalina Fiterau Brostean*		GRADUATE CMPSCI STUDENTS ONLY.	GRADUATE STUDENTS ENTERING THE COURSE ARE EXPECTED TO HAVE A STRONG WORKING KNOWLEDGE OF LINEAR ALGEBRA (MATH 545 EQUIVALENT) AND CALCULUS (EQUIVALENT TO A GRADE OF A IN MATH 233), AS WELL AS BASIC KNOWLEDGE OF PROBABILITY AND STATISTICS. SOLID PROGRAMMING SKILLS IN A HIGH-LEVEL LANGUAGE SUCH AS PYTHON, R OR MATLAB ARE EXPECTED. SEATS HELD FOR INCOMING GRAD STUDENT REGISTRATION. STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDES VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides.
COMPSCI 662 SECURTY FOR LARGE-SCALE SYSTMS *Pubali Datta*		*GRADUATE CMPSCI STUDENTS ONLY.*	*PREVIOUSLY COMPSCI 690G. SEATS SAVED FOR INCOMING GRADUATE STUDENT REGISTRATION. UNDERGRADUATE BACKGROUND IN COMPUTER SCIENCE IS ASSUMED TO INCLUDE OPERATING SYSTEMS (COMPSCI 377 OR EQUIVALENT). PROGRAMMING KNOWLEDGE AND PRINCIPLES OF COMPUTER SYSTEMS DESIGN KNOWLEDGE IS EXPECTED. STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDE VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides.*
COMPSCI 677 DISTRIBUTED AND OPERATING SYS *Peter Klemperer*		LEC 01: GRADUATE CMPSCI STUDENTS ONLY. C1 LEC 01: ONLINE MS-CMPSCI PROGRAM STUDENTS ONLY.	Lectures are available sychronously MW at 2:30-3:45 PM. Asynchronous lecture recordings are available to all students immediately after each class. All exams will be in-person for all students who are not enrolled in the online MS-CMPSCI program. Instructor permission required for senior CS majors cleared by the UPD, if available seats. Background in undergrad OS (COMPSCI 377) is helpful. Seats held for incoming grad student registration. LEC 01: Online, open to graduate CMPSCI students. C1 LEC 01: Online MS-CMPSCI program students only. Students needing special permission must request overrides via the on-line form: https://www.cics.umass.edu/academics/course-overrides.
COMPSCI 682 NEURAL NETWORKS:MODERN INTRO *Abhidip Bhattacharyya*		CMPSCI GRADUATE STUDENTS ONLY.	SEATS HELD FOR INCOMING GRAD STUDENT REGISTRATION. REQUIRED BACKGROUND FOR THE COURSE INCLUDES THE FOLLOWING. MULTIVARIATE CALCULUS, LINEAR ALGEBRA, AND PROBABILITY AND STATISTICS. PROGRAMMING IN PYTHON OR EQUIVALENT HIGH LEVEL LANGUAGE. UNDERGRADUATES AND STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDES VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides

SPIRE REGISTRATION INFORMATION for 2027 Spring

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COMPSCI 683 ARTIFICIAL INTELLIGENCE *Yair Zick*		LEC 01: GRADUATE CMPSCI STUDENTS ONLY. C1 LEC 01: ONLINE MS-CMPSCI GRADUATE STUDENTS ONLY.	STUDENTS WHO HAVE TAKEN AN UNDERGRADUATE AI COURSE PREFERRED. SEATS HELD FOR INCOMING GRAD STUDENT REGISTRATION. C1 LEC 01 FOR ONLINE MS-CMPSCI PROGRAM STUDENTS ONLY. PHDS OR ANY STUDENT WITH A TUITION WAIVER CANNOT ENROLL IN A U+ SECTION. STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDES VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides.
COMPSCI 685 ADV NATURAL LANG PROCESSING *Brendan O'Connor*		GRADUATE CMPSCI STUDENTS ONLY.	PHD-LING STUDENTS WITH PERMISSION OF INSTRUCTOR. SEATS HELD FOR INCOMING GRAD STUDENT REGISTRATION. STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDES VIA ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides
COMPSCI 688 PROBABILISTIC GRAPHICAL MODELS *Daniel Sheldon*		CMPSCI GRADUATE STUDENTS ONLY.	SEATS HELD FOR INCOMING GRAD STUDENT REGISTRATION. STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDES VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides.
COMPSCI 690AB SYSTEMS FOR DEEP LEARNING *Shiqing Ma*		CMPSCI GRADUATE STUDENTS ONLY.	STUDENTS ARE REQUIRED TO BE FAMILIAR WITH BOTH PYTHON AND C/C++ PROGRAMMING, COMPUTER SYSTEMS, AND ALGORITHMS. SOME DEGREE OF PREVIOUS FAMILIARITY WITH MACHINE LEARNING IS LIKELY TO BE NECESSARY. GRADUATE STUDENTS FROM OTHER DEPARTMENTS WITH INSTRUCTOR APPROVAL. STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDES VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides.
COMPSCI 690AG ALGORITHMIC GAME THEORY *Hedyeh Beyhaghi*		CMPSCI GRADUATE STUDENTS ONLY.	Basic background in algorithms and complexity (COMPSCI 311 or equivalent) and probability. Students needing special permission must request overrides via the on-line form: https://www.cics.umass.edu/academics/course-overrides.
COMPSCI 690CA CMPTNL SENSING:PRINC&APPLICTNS *Deepak Ganesan*		CMPSCI GRADUATE STUDENTS ONLY.	Students should have completed coursework in a graduate level Machine Learning course (e.g., COMPSCI 589, 670, or equivalent) and possess strong Python proficiency. Background in signal processing is not required but undergraduate level understanding of linear algebra and statistics is expected. Designed for PhD and advanced MS in Computer Science and Engineering. ECE Graduate Students should wait for the eligibility to open to self-enroll. Students needing special permission must request overrides via the on-line form: https://www.cics.umass.edu/academics/course-overrides.
COMPSCI 690L ADV TOPICS IN GENERATIVE AI *Chuang Gan*		C1 LEC 01: ONLINE MS-CMPSCI PROGRAM STUDENTS ONLY. LEC 01: CMPSCI GRADUATE STUDENTS ONLY.	Online synchronous participation required (Tue 10:00-12:30 EST). A strong background in probability and statistics, linear algebra, and Python programming is assumed. In addition, a background in Machine Learning (COMPSCI 589 or COMPSCI 689), Neural Networks (COMPSCI 682), and Artificial Intelligence (COMPSCI 683) is assumed. Students needing special permission must request overrides via the on-line form: https://www.cics.umass.edu/academics/course-overrides.
COMPSCI 690NN NEURAL NETWRKS-AI AND NEUROSCI *Hava Siegelmann*		GRADUATE CMPSCI STUDENTS ONLY.	Meets with COMPSCI 590NN. Seats held for incoming graduate student registration. Background in AI and Machine Learning as well as practical knowledge in Neural Networks coming from either our course (COMPSCI 682), course on LLM, or online course in this area. To accommodate students from varying backgrounds, some homework will be structured with options either requiring or not requiring programming. Students may only take this class once, either 590NN or 690NN, which are primarily differentiated through the quantity and difficulty of homework and projects, insofar as 690NN carries a higher level of academic rigor. Students needing special permission must request overrides via the on-line form: https://www.cics.umass.edu/academics/course-overrides
COMPSCI 690QC QUANTUM COMMUNICATION *Gayane Vardoyan*		CMPSCI GRADUATE STUDENTS ONLY.	MEETS WITH COMPSCI 590QC. STUDENTS CAN ONLY RECEIVE CREDIT FOR EITHER 590QC OR 690QC, NOT BOTH. SWAPPING BETWEEN 590QC AND 690QC IS ONLY CONSIDERED BEFORE THE END OF ADD/DROP. SEATS HELD FOR INCOMING GRAD STUDENT REGISTRATION. FAMILIARITY WITH PROBABILITY THEORY, LINEAR ALGEBRA, AND COMPLEX NUMBERS. NO PRIOR FAMILIARITY WITH QUANTUM CONCEPTS NEEDED. PHYSICS STUDENTS SHOULD WAIT UNTIL OPEN TO SELF-ENROLL. STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDES VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides

SPIRE REGISTRATION INFORMATION for 2027 Spring

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COMPSCI 690R CMPTNG-DIGI BIOMRKRS IN HLTHCR *Sunghoon Lee*		CMPSCI GRADUATE STUDENTS ONLY.	THERE ARE NO FORMAL PREREQUISITES FOR GRADUATE STUDENTS, BUT INFORMALLY, STUDENTS SHOULD BE FAMILIAR WITH STATISTICS AND MACHINE LEARNING. KNOWLEDGE OF HEALTH DATA OR MEDICINE IS NOT A PREREQUISITE; THIS COURSE IS DESIGNED TO INTRODUCE THE CONCEPTS OF HEALTH AND MEDICAL DATA. SEATS SAVED FOR INCOMING GRADUATE STUDENT REGISTRATION. STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDES VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides
COMPSCI 692TM S-TOPICS IN MACH LEARNG THEORY *Mingda Qiao*		CMPSCI GRADUATE STUDENTS ONLY.	PREFERRED BACKGROUND: COMPSCI 514 (OR 614) AND COMPSCI 589 (OR 689). STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDES VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides
COMPSCI 698DS P-ARTIFICIAL INTELLIGENCE *Andrew McCallum*		MS-CMPSCI GRADUATE STUDENTS ONLY.	PREVIOUSLY COMPSCI 696DS. BY THE END OF THE CURRENT FALL SEMESTER, STUDENTS MUST HAVE COMPLETED AT LEAST TWO OF THE DATA SCIENCE CORE REQUIREMENTS AND HAVE AN OVERALL GPA OF 3.0 OR HIGHER. PERMISSION OF INSTRUCTOR IS REQUIRED. MS-CMPSCI STUDENTS ARE EMAILED INSTRUCTIONS ON HOW TO APPLY FOR CONSIDERATION.
COMPSCI 701 ADV TOPICS IN CMPSCI		DOCTORAL, MASTERS GRADUATE CMPSCI STUDENTS ONLY.	THE 3 CREDIT OPTION IS FOR THE SECOND SEMESTER OF A TWO-SEMESTER SEQUENCE, 701Y FOLLOWED BY 701. THE 6 CREDIT OPTION IS FOR A PROJECT THAT WILL BE COMPLETED OVER TWO SEMESTERS WITH ENROLLMENT IN ONLY ONE SEMESTER. REQUEST OVERRIDE VIA THE SPECIAL ON-LINE FORM: https://www.cics.umass.edu/content/registration
COMPSCI 701Y ADV TOPICS IN CMPSCI (1ST SEM)		DOCTORAL, MASTERS CMPSCI GRADUATE STUDENTS ONLY.	INDICATES THE FIRST SEMESTER OF A TWO-SEMESTER SEQUENCE, 701Y (3 CREDITS) FOLLOWED BY 701 (3 CREDITS), WITH GRADE FOR BOTH ASSIGNED AT THE END. REQUEST OVERRIDE VIA THE SPECIAL ON-LINE FORM: https://www.cics.umass.edu/content/registration
COMPSCI 879 TEACH ASSTS-TOMORROW'S FACULTY *Ivon Arroyo*			COURSE IS ASSOCIATED WITH CICS TA EMPLOYMENT. REQUIRED COMPONENT OF TA CONTRACT. REGISTRATION WILL BE PROCESSED BY THE ASSOCIATE DIRECTOR GRADUATE PROGRAMS AFTER SIGNED AND APPROVED CONTRACT. OPEN TO CICS TEACHING ASSISTANTS ONLY.
COMPSCI 891M S-THEORY OF COMPUTATION *Hung Le*		DOCTORAL, MASTERS GRADUATE STUDENTS ONLY.	MAY BE REPEATED FOR CREDIT UP TO 6 TIMES. UNDERGRADUATES AND OTHER STUDENTS NOT MEETING ELIGIBILITY MUST REQUEST OVERRIDES VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides
COMPSCI 899 PHD DISSERTATION			*COURSE OPEN ONLY TO CS GRADUATE STUDENTS WHO HAVE ACHIEVED CANDIDACY. REQUEST ENROLLMENT VIA THE SPECIAL ON-LINE FORM: https://www.cics.umass.edu/academics/academic-policies/graduate-programs-policies/graduate-student-forms*
COMPSCI H230 HNRS COLLOQ: COMPUTR SYS PRINC *Meng-Chieh Chiu*	***CICS 210 (OR COMPSCI 187) WITH A GRADE OF C OR BETTER AND COMPSCI 198C [OR COMPSCI 230 WITH A GRADE OF B OR BETTER].***	*COMMONWEALTH HONORS COLLEGE SOPHOMORE AND JUNIOR CS MAJORS ONLY.*	DEPARTMENT CONSENT REQUIRED. NO OPTION FOR PASS/FAIL. STUDENTS MUST BE ENROLLED IN OR HAVE COMPLETED COMPSCI 230 WITH A GRADE OF B OR BETTER. REQUEST OVERRIDE VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides.

SPIRE REGISTRATION INFORMATION for 2027 Spring

	PREREQUISITES	ELIGIBILITY RESTRICTIONS	CLASS NOTES/COMMENTS *INFORMATION ONLY - DOES NOT AFFECT SPIRE ENROLLMENT.*
COMPSCI H311 HNRS COLLOQ: INTRO ALGORITHMS *Marius Minea*	***CICS 210 (OR COMPSCI 187) AND COMPSCI 250 (OR MATH 455) WITH A GRADE OF C OR BETTER [OR COMPSCI 311 WITH A GRADE OF B OR BETTER].***	COMMONWEALTH HONORS COLLEGE JUNIOR AND SENIOR CS MAJORS ONLY.	DEPARTMENT CONSENT REQUIRED. NO OPTION FOR PASS/FAIL. STUDENTS MUST BE ENROLLED IN COMPSCI 311 (OR HAVE COMPLETED COMPSCI 311 WITH A GRADE OF B OR BETTER). MEETING SHALL BE MONDAYS AT TO BE DETERMINED TIME BY INSTRUCTOR AT START OF SEMESTER. INTERESTED STUDENTS MUST REQUEST OVERRIDE VIA ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides.
COMPSCI H377 HNRS COLLOQ: OPERATING SYSTEMS *Phuthipong Bovornkeeratiroj*	***COMPSCI 230 WITH A GRADE OF C OR BETTER [OR COMPSCI 377 WITH A GRADE OF B OR BETTER].***	*COMMONWEALTH HONORS COLLEGE JUNIOR AND SENIOR CS MAJORS ONLY.*	*NO OPTION FOR PASS/FAIL. STUDENTS MUST BE ENROLLED IN COMPSCI 377 (OR HAVE COMPLETED COMPSCI 377 WITH A GRADE OF B OR BETTER). REQUEST OVERRIDE VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides.*
COMPSCI H589 HNRS COLLOQ: MACHINE LEARNING	***[MATH 545 AND STATISTCS 315 (PREVIOUSLY STATISTCS 515) AND COMPSCI 240, ALL WITH A GRADE OF C OR BETTER.] OR [MATH 545 WITH A GRADE OF C OR BETTER AND COMPSCI 240 WITH A GRADE OF B+ OR BETTER.] OR [MATH 235 AND MATH 233, BOTH WITH A GRADE OF B+ OR BETTER, AND STATISTICS 315 (PREVIOUSLY STATISTCS 515) AND COMPSCI 240, BOTH WITH A GRADE OF C OR BETTER.] OR [MATH 235 AND MATH 233 AND COMPSCI 240, ALL WITH A GRADE OF B+ OR BETTER.] OR [COMPSCI 589 WITH A GRADE OF B OR BETTER.]***	COMMONWEALTH HONORS COLLEGE JUNIOR AND SENIOR CS MAJORS ONLY.	*NO OPTION FOR PASS/FAIL. DEPARTMENT CONSENT REQUIRED. STUDENTS MUST BE ENROLLED IN COMPSCI 589 (OR HAVE COMPLETED COMPSCI 589 WITH A GRADE OF B OR BETTER). REQUEST OVERRIDE VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides*
INFO 101 INTRO TO INFORMATICS *Justin Clarke*		UNDERGRADUATE NON-CS MAJORS ONLY.	CS MAJORS ARE NOT ELIGIBLE FOR THIS COURSE. FOUNDATIONS COURSE FOR THE INFORMATICS MAJOR. STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDES VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides.
INFO 150 MATH FOUND-INFORMATICS *STAFF*		INFORM MAJORS ONLY.	CS AND MATH (STATISTICS) MAJORS ARE NOT ELIGIBLE FOR THIS COURSE. AFTER INFORM MAJORS REGISTER, INFO 150 WILL OPEN TO ALL NON-CS AND NON-MATH/STATS UNDERGRADUATES WHO SHOULD WAIT FOR THE MAJOR ELIGIBILITY RESTRICTION TO BE REMOVED TO SELF-ENROLL OR ADD TO THE WAITLIST (IF AVAILABLE) AND SHOULD NOT REQUEST AN OVERRIDE. ANTICIPATED OPENING DURING THE FIRST WEEK OF CLASSES. R1 MATH USEFUL; NOT ABSOLUTELY REQUIRED. STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDES VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides
INFO 203 A NETWORKED WORLD *Mohammadhassan Hajiesmaili*		INFORM MAJORS ONLY.	CS MAJORS ARE NOT ELIGIBLE FOR THIS COURSE. AFTER INFORM MAJORS REGISTER, INFO 203 WILL OPEN TO ALL NON-CS UNDERGRADUATES WHO SHOULD WAIT FOR THE MAJOR ELIGIBILITY RESTRICTION TO BE REMOVED TO SELF-ENROLL OR ADD TO THE WAITLIST (IF AVAILABLE) AND SHOULD NOT REQUEST AN OVERRIDE. ANTICIPATED OPENING DURING THE FIRST WEEK OF CLASSES. STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDES VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides
INFO 248 INTRO TO DATA SCIENCE *Gordon Anderson*	CICS 110 (OR CICS 160 OR COMPSCI 119 OR COMPSCI 121) WITH A GRADE OF C OR BETTER, AND EITHER PSYCH 240, OIM 240, STATISTC 240, RES-ECON 212, SOCIOL 212, OR STATISTC 315/515, OR COMPSCI 240, WITH A GRADE OF C OR BETTER.	INFORM MAJORS ONLY.	PREVIOUS EXPERIENCE WITH A PROGRAMMING LANGUAGE (OTHER THAN COMPSCI 121) MAY BE CONSIDERED FOR OVERRIDE. BASIC MATHEMATICAL MATURITY (COLLEGE ALGEBRA) IS ASSUMED. BASIC PROBABILITY AND STATISTICS, SUCH AS ONE OF THE COURSES REQUIRED FOR REGISTRATION ON SPIRE IS REQUIRED. THIS COURSE CONSISTS OF TWO LECTURES AND ONE LAB SECTION PER WEEK. EACH LECTURE MEETING INCLUDES A PRESENTATION OF CONCEPTUAL MATERIAL AS WELL AS SOME TIME TO WORK ON IMPLEMENTATION OF THE MATERIAL PRESENTED. IN LAB TIME, STUDENTS RECEIVE DETAILED INSTRUCTION ON THE USE OF SOFTWARE TOOLS TO ANALYZE ACTUAL DATA SETS. READINGS WILL BE ASSIGNED AS PREPARATION FOR EACH CLASS MEETING. A PROJECT WILL BE ASSIGNED DURING THE COURSE. STUDENTS WORK IN PAIRS ON THE PROJECT. THE PROJECT PROVIDES STUDENTS WITH AN OPPORTUNITY TO WORK COLLABORATIVELY TO EXPLORE A TOPIC IN MORE DEPTH. AFTER ELIGIBLE INFORM MAJORS REGISTER, WILL OPEN TO ALL ELIGIBLE UNDERGRADUATES, IF AVAILABLE SEATS. ANTICIPATED OPENING DURING THE FIRST WEEK OF CLASSES. STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDES VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides

SPIRE REGISTRATION INFORMATION for 2027 Spring

	PREREQUISITES	ELIGIBILITY RESTRICTIONS	CLASS NOTES/COMMENTS *INFORMATION ONLY - DOES NOT AFFECT SPIRE ENROLLMENT.*
INFO 348 DATA ANALYTICS WITH PYTHON *Jonathan Samuel Stern*	INFO 248 AND CICS 160 (OR INFO 190T OR COMPSCI 186 OR COMPSCI 187), BOTH WITH A GRADE OF C OR BETTER.	INFORM MAJORS ONLY.	PREVIOUSLY CICS 397A. LINEAR ALGEBRA (MATH 235) IS HELPFUL BUT NOT REQUIRED. STUDENTS WITH ALTERNATE PROGRAMMING OR STATISTICS BACKGROUND MAY REQUEST OVERRIDE FOR CONSIDERATION. A PREREQUISITE CHANGE HAS BEEN APPROVED BY THE FACULTY SENATE TO REPLACE CICS 210 WITH CICS 160. AFTER INFORM MAJORS REGISTER, INFO 348 WILL OPEN TO ALL UNDERGRADUATES, IF AVAILABLE SEATS. STUDENTS MEETING PREREQUISITE SHOULD WAIT FOR THE MAJOR ELIGIBILITY RESTRICTION TO BE REMOVED TO SELF-ENROLL OR ADD TO THE WAITLIST (IF AVAILABLE) AND SHOULD NOT REQUEST AN OVERRIDE. STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDES VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides
INFO 494PI PERSONAL HEALTH INFORMATICS *Ravi Karkar*	*INFO 248 (OR COMPSCI 240) AND CICS 210 (OR COMPSCI 186 OR COMPSCI 187) BOTH WITH A GRADE OF C OR BETTER.*	JUNIOR AND SENIOR INFORM MAJORS ONLY.	*NO PASS/FAIL OPTION. SATISFIES INTEGRATIVE EXPERIENCE REQUIREMENT FOR CS AND INFORM MAJORS. ALSO SATISFIES A CS400 ELECTIVE FOR CS MAJORS. WILL OPEN TO JUNIOR AND SENIOR CS MAJORS AFTER JUNIOR AND SENIOR INFORM MAJORS ARE GIVEN OPPORTUNITY TO ENROLL, IF AVAILABLE SEATS. ELIGIBLE CS MAJORS SHOULD NOT REQUEST AN OVERRIDE, BUT WAIT FOR ELIGIBILITY TO OPEN TO SELF-ENROLL. STUDENTS NEEDING SPECIAL PERMISSION MUST REQUEST OVERRIDES VIA THE ON-LINE FORM: https://www.cics.umass.edu/academics/course-overrides*
INFO H494PI HNRS COLLOQ: PERS HLTH INFORM *Ravi Karkar*	***INFO 248 AND (COMPSCI 186 OR COMPSCI 187 OR CICS 210), (OR INFO 494PI WITH A GRADE OF B OR BETTER***).	JUNIOR AND SENIOR CS AND INFORM MAJORS IN COMMONWEALTH HONORS COLLEGE.	*NO OPTION FOR PASS/FAIL. STUDENTS MUST BE ENROLLED IN INFO 490PI (OR HAVE COMPLETED INFO 490PI WITH A GRADE OF B OR BETTER). REQUEST OVERRIDE VIA THE ON-LINE FORM: https://www.cics.umass.edu/overrides*